

Machine Learning Assignment 1

Polynomial Regression — Steam Turbine Optimisation & Thermal Reservoir Mapping

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Contents

1	Problem Overview	2
2	Methodology	2
2.1	General Pipeline	2
2.2	Polynomial Feature Expansion	2
2.3	Regularisation Strategy	2
3	Phase 1 – Steam Turbine Optimisation (var1)	2
3.1	Dataset	2
3.2	Degree Selection	3
3.3	Analysis	3
4	Phase 2 – Thermal Reservoir Mapping (var2)	3
4.1	Dataset	3
4.2	Degree Selection	4
4.3	Analysis	4
5	Summary	5
6	Submission Files	5

1 Problem Overview

Two polynomial regression problems are addressed, each with a personalised dataset.

- **var1 – Steam Turbine Optimisation:** Predict the Net Power Score y from six operational parameters $(x_1, \dots, x_6)$ using a polynomial of degree up to 10.
- **var2 – Thermal Reservoir Mapping:** Predict the Thermal Anomaly Score y from three spatial coordinates (x_1, x_2, x_3) using a polynomial of degree up to 20.

Both datasets are pre-cleaned. The target variable in every file is y .

2 Methodology

2.1 General Pipeline

For both problems the same five-step pipeline is applied:

1. Split the training data into an 80% development set and a 20% hold-out set (fixed `random_state=42`).
2. For each candidate degree, fit a regularised polynomial model on the development set using 5-fold cross-validation to tune the regularisation strength. Record the mean CV MSE across folds at the best regularisation parameter.
3. Select the degree with the lowest CV MSE.
4. Evaluate the chosen degree *once* on the untouched hold-out set (MSE, R^2).
5. Refit on the full training data with the chosen degree and produce predictions for the test set.

2.2 Polynomial Feature Expansion

Given input $\mathbf{x} \in R^p$, polynomial features of degree d generate all monomials $x_1^{a_1} x_2^{a_2} \dots x_p^{a_p}$ with $\sum_i a_i \leq d$. The expanded features are standardised (zero mean, unit variance) before fitting.

2.3 Regularisation Strategy

var1 — Lasso (L1). With $p = 6$ features and degree up to 10, the expanded feature space can exceed several thousand terms. Lasso’s L1 penalty drives irrelevant coefficients to exactly zero, performing automatic feature selection and preventing overfitting. The regularisation parameter α is chosen by `LassoCV` over a log-spaced grid of 20 values.

var2 — Ridge (L2). With only $p = 3$ features the polynomial expansion remains manageable even at degree 20. Ridge’s L2 penalty shrinks all coefficients smoothly without zeroing them, which suits the smooth geological surface being modelled. The regularisation parameter α is chosen by `RidgeCV` over a log-spaced grid of 40 values ranging from 10^{-6} to 10^4 .

3 Phase 1 – Steam Turbine Optimisation (var1)

3.1 Dataset

Training: 1000 samples, 6 features. Test: 1000 samples.

3.2 Degree Selection

Table 1: var1: CV MSE and hold-out metrics for each polynomial degree (Lasso).

Degree	CV MSE	Hold-out MSE	Hold-out R^2
1	9.0011	8.1066	0.1166
2	3.1111	2.7741	0.6977
3	1.0760	0.9165	0.9001
4	0.7242	0.6029	0.9343
5	0.3467	0.3590	0.9609
6	0.3679	0.3637	0.9604
7	0.3865	0.3818	0.9584
8	0.3941	0.3758	0.9591
9	0.3994	0.3788	0.9587
10	0.4037	0.3816	0.9584

3.3 Analysis

The CV MSE decreases steeply from degree 1 to 5 and then rises monotonically from degree 6 onward (Figure 1), indicating overfitting beyond degree 5. **Degree 5** is selected as the optimal degree.

On the held-out 20%:

$$\text{MSE} = 0.3590, \quad R^2 = 0.9609$$

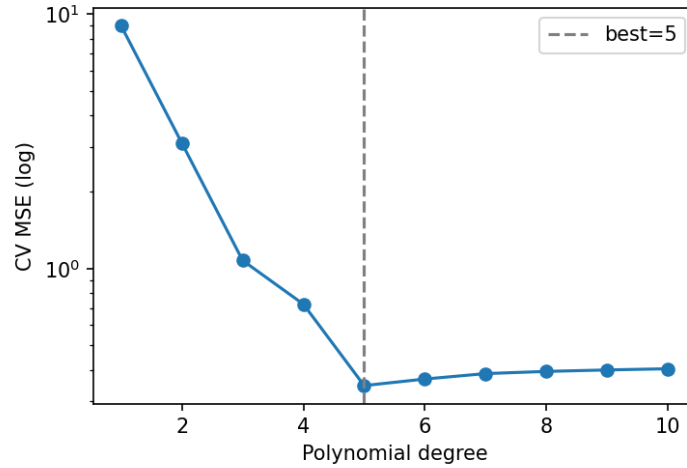


Figure 1: var1: CV MSE vs. polynomial degree. The dashed line marks degree 5.

4 Phase 2 – Thermal Reservoir Mapping (var2)

4.1 Dataset

Training: 1000 samples, 3 features. Test: 1000 samples.

4.2 Degree Selection

Table 2: var2: CV MSE and hold-out metrics for each polynomial degree (Ridge).

Degree	CV MSE	Hold-out MSE	Hold-out R^2
1	39.4143	25.1706	0.2779
2	23.4251	17.8965	0.4866
3	12.9633	8.6325	0.7523
4	4.2470	2.9721	0.9147
5	1.7873	1.2768	0.9634
6	0.6090	0.5482	0.9843
7	0.3764	0.3686	0.9894
8	0.2618	0.3108	0.9911
9	0.2677	0.3013	0.9914
10	0.2534	0.2905	0.9917
11	0.2600	0.2923	0.9916
12	0.2592	0.2907	0.9917
13	0.2671	0.2954	0.9915
14	0.2737	0.2936	0.9916
15	0.2807	0.2993	0.9914
16	0.2904	0.3047	0.9913
17	0.2965	0.3106	0.9911
18	0.3074	0.3135	0.9910
19	0.3139	0.3186	0.9909
20	0.3277	0.3227	0.9907

4.3 Analysis

CV MSE decreases from degree 1 to 10 and begins to increase from degree 11 (Figure 2), consistent with a complex but smooth thermal surface. **Degree 10** is selected as the optimal degree.

On the held-out 20%:

$$\text{MSE} = 0.2905, \quad R^2 = 0.9917$$

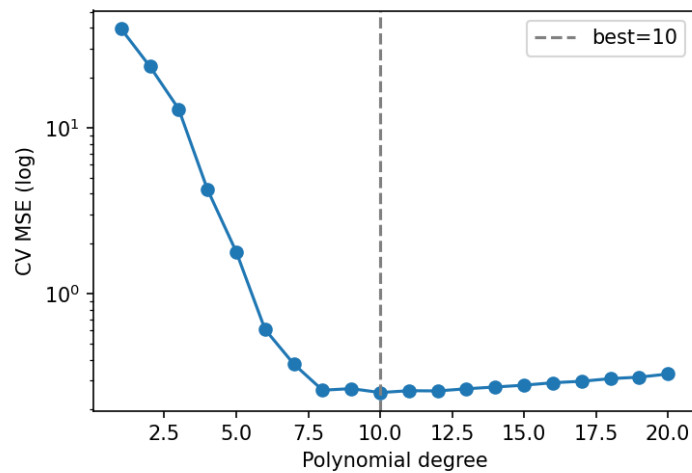


Figure 2: var2: CV MSE vs. polynomial degree. The dashed line marks degree 10.

5 Summary

Table 3: Final model summary.

Problem	Regulariser	Best Degree	Hold-out MSE	Hold-out R^2
var1 (Steam Turbine)	Lasso	5	0.3590	0.9609
var2 (Thermal Reservoir)	Ridge	10	0.2905	0.9917

Both models achieve $R^2 > 0.96$, indicating strong generalisation on the hold-out data. The degree selection was driven entirely by cross-validated MSE with no look-ahead into the test set.

6 Submission Files

- `BT2024218_pred_var1.csv` — 1000 predicted Net Power Scores.
- `BT2024218_pred_var2.csv` — 1000 predicted Thermal Anomaly Scores.
- `part1.py` — training and inference code for var1.
- `part2.py` — training and inference code for var2.